Project Name	TaskOrbit Conversational Agent
Online team meeting	https://fau.zoom-x.de/j/65288966243?pwd=DnF4jxSAfJugyMmynzjE1BaaFprRbi.1
Production system (if any)	GCP
Test system (if any)	TBD (Industry Partner Meeting on Monday)
GitHub repository	https://github.com/amosproj/amos2026ss04-taskorbit-conversational-agent
GitHub feature board	https://github.com/orgs/amosproj/projects/100/views/2
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/104/views/1
Position of T-shirt logo	Upper Torso Center Large
Link to logo for black T-shirt	https://github.com/amosproj/amos2026ss04-taskorbit-conversational-agent/blob/feature/team-logo-design/Deliverables/sprint-01/team-logo
Link to logo for white T-shirt	https://github.com/amosproj/amos2026ss04-taskorbit-conversational-agent/blob/feature/team-logo-design/Deliverables/sprint-01/team-logo
Additional materials	https://discord.gg/usXpc5ab
Team maling list	oss-amos-proj4@lists.fau.de

Last Name	First Name	GitHub User Name	Email Address
Fatima	Qiblatain	qiblatainf	qiblatain.fatima@fau.de
Huy	Christoph	Perimora	christoph.huy@campus.tu-berlin.de
Raza	Asad	asadraza17	asad.raza@fau.de
Jokiel	Rene	BelmontR	rene.jokiel@fau.de
Mosler	Carl	Caerum5	carl.mosler@tu-berlin.de
Vineesh	Koppay	K-Vineesh	vineesh.koppay@fau.de
Moeez	Abdul	abdulmoeez1225	abdul.moeez@fau.de
Thakur	Shikhar	shikharthakur2404`	shikhar.thakur@fau.de
Dhruvin	Vekariya	djv03	dhruvin.vekariya@fau.de
Shameti	Endri	1Endri	endri.shameti@fau.de

		Product Owner					
#	**Meeting Day**	**Review**	**Planning**	**Software Developer**	**Release Manager**	**Scrum Master**	**Comment**
1	2026-04-15		Carl	Everyone else	N/A	Rene	
2	2026-04-22	Carl	Christoph		Endri Shameti	Rene	
3	2026-04-29	Christoph	Carl		Dhruvin	Rene	
4	2026-05-06	Carl	Christoph		Qiblatain	Rene	
5	2026-05-13	Christoph	Carl		Vineesh Koppay	Rene	
6	2026-05-20	Carl	Christoph		Asad Raza	Rene	
7	2026-05-27	Christoph	Carl		Abdul Moeez	Rene	Mid-term due
8	2026-06-03	Carl	Christoph		Shikhar	Rene	
9	2026-06-10	Christoph	Carl		Endri Shameti	Rene	
10	2026-06-17	Carl	Christoph		Dhruvin	Rene	
11	2026-06-24	Christoph	Carl		Qiblatain	Rene	
12	2026-07-01	Carl	Christoph		Vineesh Koppay	Rene	
13	2026-07-08	Christoph	Carl		Abdul Moeez	Rene	
14	2026-07-15	Carl	Christoph		Asad Raza	Rene	Demo day!
15	2026-07-22	Christoph	Carl		Shikhar	Rene	Retrospective
Product owners, software developers, and Scurm Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Create a flawless and valuable project that the industry partner will be able to use. Create clean and reusable code and structures
Meeting norms	- be on time -cams on -reach out if you are not gonna make it
	- We'll respect each other's opinions and try to communicate as much as possible
Working norms	- Document and comment your code - We'll use branches in a sensefull way and not work directly on the main branch - Transparency - Clear commit messages - We test our code - We write clean code
Coordination norms	- We'll update the feature board in time - If there are multiple devs working on one item, they will communicate and coordinate and not work independently - We balance the workload among the team
Communication norms	- We talk about issues and problems when we face them - Transparent communication - Daily check the corresponding channels
Consideration norms	- We support each other when ever needed - We talk about issues openly - If we can't reach a consensus, we'll vote - We don't take critique personally
Cont. improvement norms	- We'll be open to learn and teach - We encourage critique and improvements efforts
Rewards	- We celebrate milestones with pizza - Very helpful team members get an extra slice of pizza
Sanctions	- Sit ups (at least 10, more if needed)
Signatures	
Scrum Master	Rene Jokiel
Product owner	Christoph Huy
Product owner	Carl Mosler
Software developer	Endri Shameti
Software developer	Dhruvin Vekariya
Software developer	Abdul Moeez
Software developer	Vineesh Koppay
Software developer	Asad Raza
Software developer	Qiblatain Fatima
Software developer	Shikhar Thakur

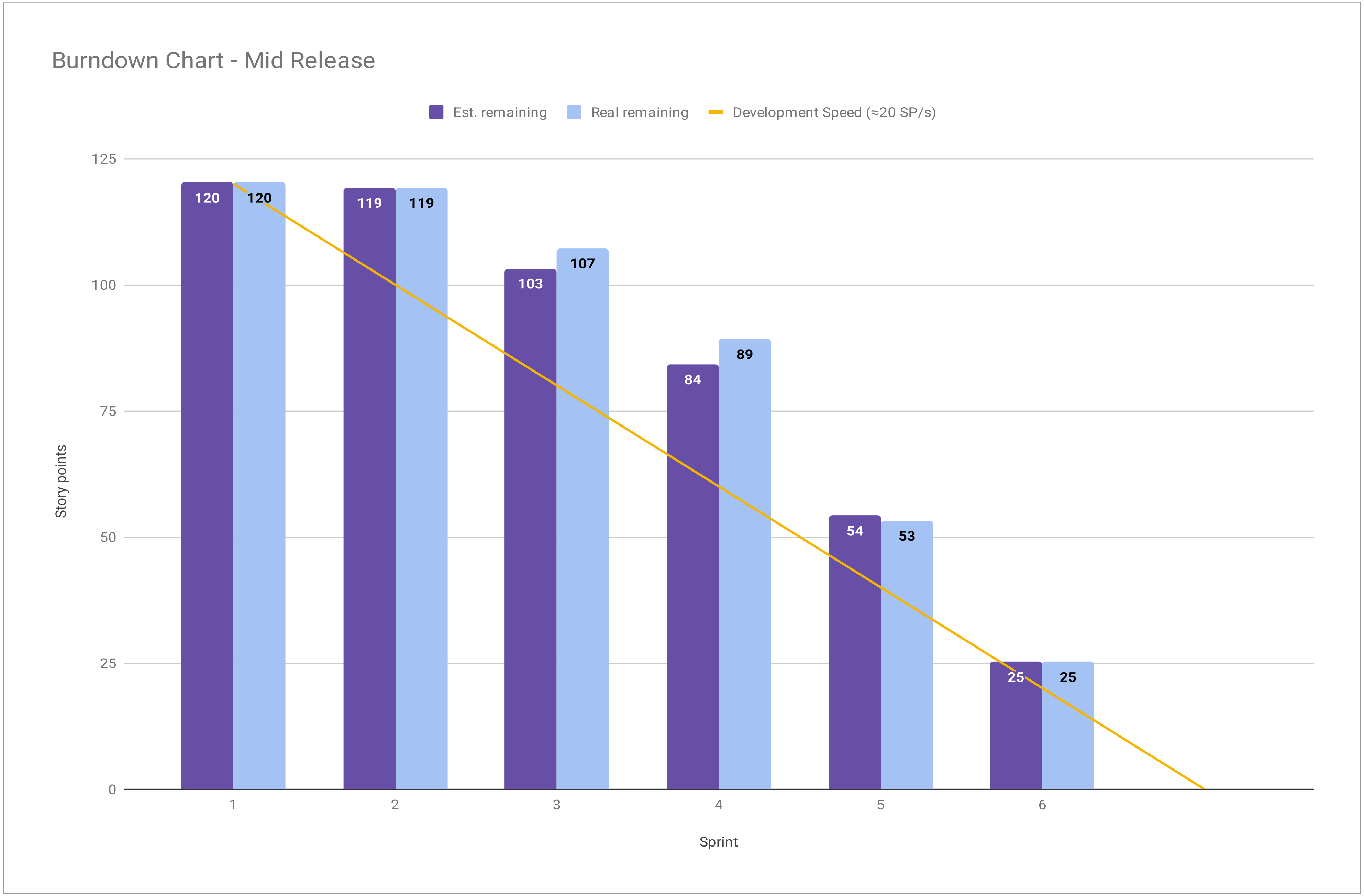
Product Vision	Project Mission
Product Vision: TaskOrbit provides a domain-independent framework for building reliable conversational AI agents through declarative workflows and structured agent orchestration. The platform enables organizations to create configurable voice and chat agents that integrate STT, LLMs, TTS, and external APIs into unified conversational workflows. By standardizing conversational automation across different business domains, TaskOrbit helps companies reduce manual effort, streamline internal and external processes, and deploy scalable AI-driven assistants without building custom infrastructure from scratch.	The mission of this project is to create an MVP for a Voice AI Agent platform. Within the given project time-frame, the team is committed to delivering a fully functional conversational interface supporting configurable single-agent and multi-agent workflows. Core functionality will include the integration of STT for user input, a LLM backend for conversational processing and reasoning, agent orchestration capabilities, and TTS for audio output. The primary objective of this project is to provide a reliable and extensible end-to-end conversational system through the integration of these core components.

Term	Definition
E2E	End-to-End (Workflow): The complete processing pipeline of a system interaction, starting from user input through internal processing and concluding with the system’s resulting output.
LLM (Large Language Model)	The underlying AI backend used for conversational processing, reasoning, and generating text responses based on user prompts
TTS (Text to Speech)	The technology that converts text into a spoken audio output (e.g. the voice of the agent talking to the user)
STT (Speech to Text)	The technology that converts spoken audio input into a text (e.g., transcribing the spoken input from a user for a LLM)
Latency	The time delay between the end of a user's spoken input and the beginning of the agent's audio response (Roundtrip Time)
Agent	An autonomous entity within the platform configured with a specific persona, logic and context to interact with the user via voice (or text)
Agent Orchestration	The capability of the TaskOrbit framework to coordinate multiple specialized conversational agents within a single workflow to solve tasks
Tool	An external integration, API or function invoked by the agent to interact with outside services, databases or systems
Skill	An internal capability, native logic or workflow executed within the agent's system to process information or manage tasks

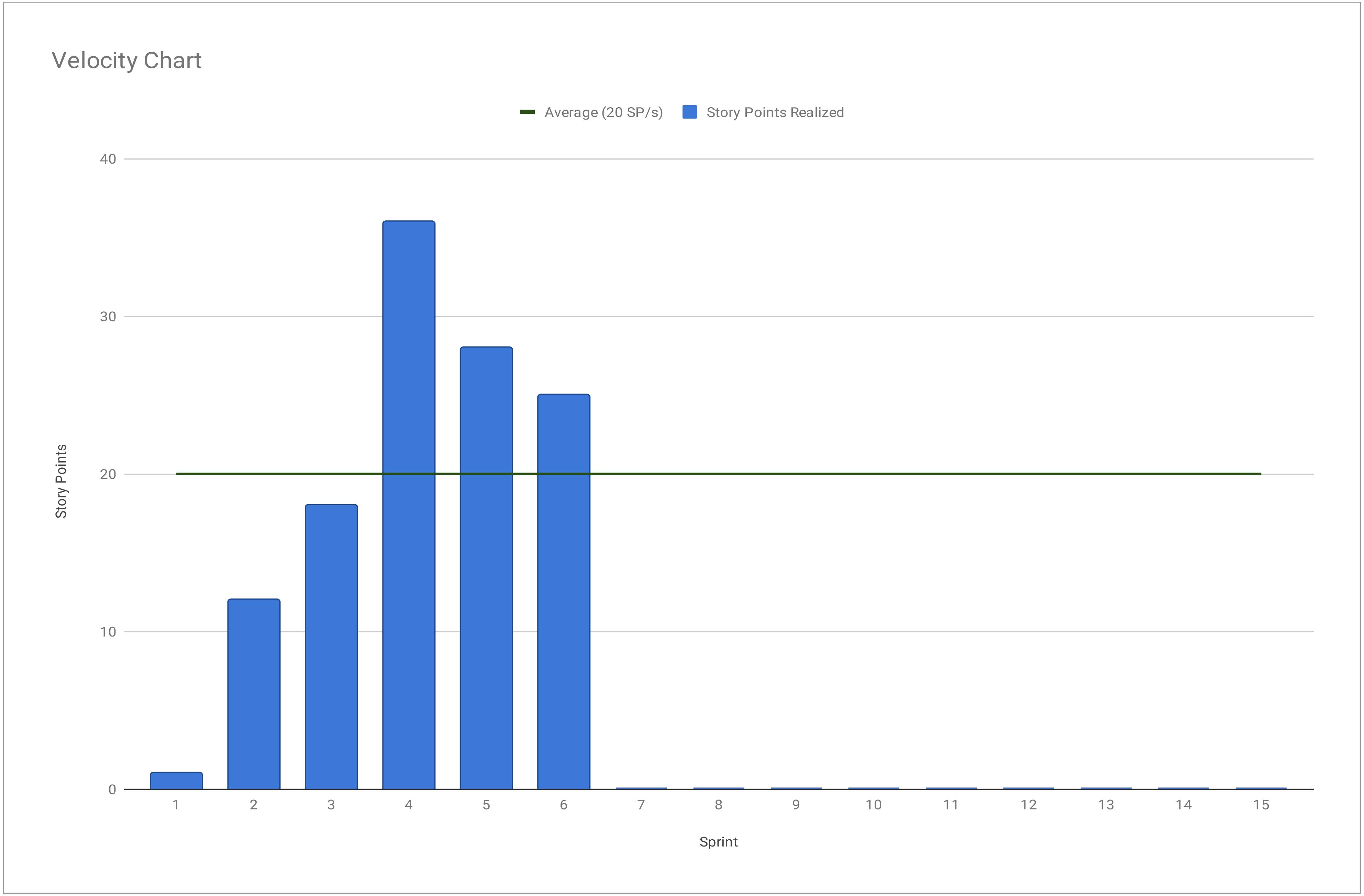
Sprint #	Sprint goal
1	None
2	None
3	None
4	Optional
5	Persistent Single-Prompt Agent Prototype
6	Cloud Deployment/Migration & System Robustness
7	Advance core platform capabilities & foundation for multi-agent workflow orchestration
8	
9	
10	
11	
12	
13	
14	
15	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			120	120		
Sprints						
1			1	120	1	120
2			16	119	12	119
3			19	103	18	107
4			30	84	36	89
5			29	54	28	53
6			25	25	25	25
Features						
1	**None**	Design Team Logo #1	1		1	
2	**None**	Initialize Software Bill of Materials #13	3		2	
		System Architecture #12	5		5	
		First UI Design & Implementation #5	5		5	
		Setup Development Environment #17	3		Not Accepted	
3	**None**	Setup Development Environment #17	3		3	
		Setup LiveKit Infrastructure #26	5		5	
		Schema Foundation #6	3		3	
		Prepare Build Process Review #22	3		2	
		Core Backend Foundation #7	5		5	
4	**Optional**	CI/CD Pipeline #10	5		8	
		Basic Agent Runtime #33	5		5	
		Basic LLM Integration #18	5		5	
		DB Integration #32	5		5	
		Audio Output via LiveKit (Text-to-Speech) # 15	5		5	
		Audio Input via LiveKit (Speech-to-Text) #14	5		8	
5	**Persistent Single-Prompt Agent Prototype**	Build Process Video #43	2		3	
		User Conversation History #45	5		Not Accepted	
		Save Agent Configuration #47	3		3	
		Load Agent Configuration #51	3		3	
		Manage Agent Configuration #52	3		3	
		Logging & Monitoring Setup #11	8		8	
		Voice Interruption for Audio Chat #55	5		8	
6	**Cloud Deployment/Migration & System Robustness**	User Conversation History #45	5		25	
		User, Design & Build Documentation #74	2			
		Slot Extraction for Structured Task Inputs #70	5			

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Integrate Cloud Environment with IaC #21	8			
		Agent Persona Guardrails #69	5			
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				


Burndown Chart - Mid Release
Est. remaining
Real remaining
Development Speed (≈20 SP/s)
125
100
75
50
25
0
Story points
120
120
119
119
103
107
84
89
54
53
25
25
1
2
3
4
5
6
Sprint

Sprint #	Story Points Realized
1	1
2	12
3	18
4	36
5	28
6	25
7	
8	
9	
10	
11	
12	
13	
14	
15	
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB


Velocity Chart
Average (20 SP/s)
Story Points Realized
40
30
20
10
0
Story Points
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Sprint

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
7			0	0	0	0
8			0	0	0	0
9			0	0	0	0
10				0		0
11						
12						
13						
Features						
7	Advance core platform capabilities & foundation for multi-agent workflow orchestration					
8						
9						

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done
	General coding standards are met (eg. Formatting, Comments, Logging, no API-Keys in Repo)		
	Every PR is peer-reviewed by at least one other developer		
	Documentation is updated (eg. System Architecture, SBOM)		
	Unit tests for new features have been written		
		A release candidate representing a functional and verifiable increment over the previous sprint is tagged	
		Existing features must remain operational and unaffected by the new changes	
		Sprint release changelog have been written	
			The system can be built and started from a clean repository checkout
			Core workflows are fully integrated and demonstrable end-to-end
			All major components are integrated and operational
			No critical or blocker defects remain open
			Architecture and technical documentation are finalized and updated
			SBOM and required project documentation are completed
			CI/CD pipeline passes successfully for the release version
			Release version is tagged and reproducible
			Build and deployment instructions are documented and verified
			Source code, documentation, and deliverables are submitted

Type	Link / reference
Team Logo (SVG + PNG)	https://github.com/amosproj/amos2026ss04-taskorbit-conversational-agent/blob/feature/team-logo-design/Deliverables/sprint-01/team-logo

#	Context	Name	Version	License	Comment
1	python	fastapi	^0.115.0	MIT	REST API framework
2	python	uvicorn	^0.32.0	BSD	ASGI server
3	python	pydantic	^2.9.0	MIT	Data validation
4	python	pydantic-settings	^2.6.0	MIT	Settings management
5	python	python-dotenv	^1.0.1	BSD	Environment config
6	python	livekit-agents	^1.5.0	Apache-2.0	Voice agent framework
7	python	sqlalchemy	^2.0.36	MIT	Database ORM
8	python	alembic	^1.14.0	MIT	Database migrations
9	python	psycopg	^3.2.0	LGPL	PostgreSQL driver
10	python	structlog	^24.4.0	MIT	Structured logging
11	python	prometheus-fastapi-instrumentator	^7.0.0	ISC	FastAPI middleware for Prometheus metrics
12	python	prometheus-client	^0.22.0	Apache-2.0	Official Prometheus instrumentation client
13	python	opentelemetry-api	>=1.27.0	Apache-2.0	OpenTelemetry tracing and metrics API
14	python	opentelemetry-sdk	>=1.27.0	Apache-2.0	OpenTelemetry reference implementation SDK
15	python	httpx	^0.28.1	BSD	HTTP client
16	npm	react	^19.0.0	MIT	UI framework
17	npm	react-dom	^19.0.0	MIT	React DOM renderer
18	npm	livekit-client	^2.5.0	Apache-2.0	LiveKit client SDK
19	npm	@livekit/components-react	^2.6.0	Apache-2.0	LiveKit React components
20	npm	@livekit/components-styles	^1.1.0	Apache-2.0	LiveKit component styles
21	npm	react-router-dom	^7.1.0	MIT	Client-side routing
22	npm	sonner	^2.0.7	MIT	Toast notifications
23	npm	next-themes	^0.4.6	MIT	Theme management
24	npm	@radix-ui/react-slot	^1.1.0	MIT	Slot component for Radix UI
25	npm	class-variance-authority	^0.7.0	Apache-2.0	Variant management for classes
26	npm	clsx	^2.1.1	MIT	Class name utility
27	npm	lucide-react	^0.460.0	ISC	Icon library for React
28	npm	radix-ui	^1.4.3	MIT	Radix UI primitives
29	npm	tailwind-merge	^2.5.0	MIT	Tailwind class merging
30	python	aiosqlite	^0.22.1	MIT	Async SQLite driver for testing

#	Context	Name	Version	License	Comment

Last Name	First Name	Value					
Fatima	Qiblatain	0		0.00	OK		
Huy	Christoph	0					
Raza	Asad						
Jokiel	Rene						
Mosler	Carl			0	No size		
Vineesh	Koppay			1	Trivial size		
Moeez	Abdul			2	Small size		
Thakur	Shikhar			3	Medium size		
Dhruvin	Vekariya			5	Large size		
				8	Very large size		
				13	Too large (size)		

How to play planning poker

1. Everyone type their number into their value field, don't hit return yet
2. Someone, perhaps a product owner, count down 3.. 2.. 1..
3. Then, everyone hit return to submit their value

Sprint #	Story Points Realized	Average (20 SP/s)
1	1	20
2	12	20
3	18	20
4	36	20
5	28	20
6	25	20
7		20
8		20
9		20
10		20
11		20
12		20
13		20
14		20
15		20

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining	Development Speed
Release							
Total			120	120			20
Sprints							
							Development Speed (≈20 SP/s)
1			1	120	1	120	120
2			16	119	12	119	100
3			19	103	18	107	80
4			30	84	36	89	60
5			29	54	28	53	40
6			25	25	25	25	20
							0